Jaejun Lee (이재준, 李在埈)

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Education

KAIST *Daejeon*

Ph.D., School of Computing *2023.02 - present*

- Big Data Intelligence Lab (KAIST BDI Lab)
- Advisor: Professor Joyce Jiyoung Whang

KAIST *Daejeon*

M.S., School of Computing *2021.08 - 2023.02*

- Big Data Intelligence Lab (KAIST BDI Lab)
- Advisor: Professor Joyce Jiyoung Whang

KAIST *Daejeon*

B.S., School of Computing *2017.02 - 2021.08*

- Double Major: Department of Mathematical Sciences

Publications

† *equal contribution*, ⋆ *corresponding author*

1. **Jaejun Lee** and Joyce Jiyoung Whang⋆, "Learn When and Where to Connect: Adaptive Virtual Nodes for Dynamic Message Passing on Graphs", *Proceedings of the 32nd SIGKDD Conference on Knowledge Discovery and Data Mining V.2* (*KDD*), pages 2485-2496, August 2026.
2. **Jaejun Lee**, Seheon Kim, and Joyce Jiyoung Whang⋆, "Generative Representation Learning on Hyper-relational Knowledge Graphs via Masked Discrete Diffusion", *Proceedings of the 43rd International Conference on Machine Learning* (*ICML*), pages 64205-64232, July 2026.
3. **Jaejun Lee** and Joyce Jiyoung Whang⋆, "Structure Is All You Need: Structural Representation Learning on Hyper-Relational Knowledge Graphs", *Proceedings of the 42nd International Conference on Machine Learning* (*ICML*), pages 33765-33783, July 2025.
4. Minsung Hwang, **Jaejun Lee**, and Joyce Jiyoung Whang⋆, "Stability and Generalization Capability of Subgraph Reasoning Models for Inductive Knowledge Graph Completion", *Proceedings of the 42nd International Conference on Machine Learning* (*ICML*), pages 26376-26411, July 2025.
5. **Jaejun Lee**, Minsung Hwang, and Joyce Jiyoung Whang⋆, "PAC-Bayesian Generalization Bounds for Knowledge Graph Representation Learning", *Proceedings of the 41st International Conference on Machine Learning* (*ICML*), pages 26589-26620, July 2024.
6. **Jaejun Lee**, Chanyoung Chung, Hochang Lee, Sungho Jo, and Joyce Jiyoung Whang⋆, "VISTA: Visual-Textual Knowledge Graph Representation Learning", *Findings of the Association for Computational Linguistics*: *EMNLP 2023* (*Findings of EMNLP*), pages 7314-7328, December 2023.
7. Chanyoung Chung†, **Jaejun Lee**†, and Joyce Jiyoung Whang⋆, "Representation Learning on Hyper-Relational and Numeric Knowledge Graphs with Transformers", *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining* (*KDD*), pages 310-322, August 2023.
8. **Jaejun Lee**, Chanyoung Chung, and Joyce Jiyoung Whang⋆, "InGram: Inductive Knowledge Graph Embedding via Relation Graphs", *Proceedings of the 40th International Conference on Machine Learning* (*ICML*), pages 18796-18809, July 2023.
9. Ji Ho Kwak†, **Jaejun Lee**†, Sungho Jo⋆, and Joyce Jiyoung Whang⋆, "Semantic Grasping via a Knowledge Graph of Robotic Manipulation: A Graph Representation Learning Approach", *IEEE Robotics and*

Automation Letters, Vol. 7, No. 4, pages 9397-9404, October 2022. Also presented at *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems* (*IROS*)

Teaching Experience

Tutorials

- Assistant Tutor of "Knowledge Discovery with Knowledge Graphs: From Structural Embeddings to Generative Reasoning(KnowKG)", To Appear in *SIGKDD Conference on Knowledge Discovery and Data Mining*(*KDD*), Tutorial, August 2026.
- Assistant Tutor of "Key Facets in Modern Knowledge Graph Representation Learning (KeyKGRL)", *International Semantic Web Conference* (*ISWC*), Tutorial, November 2025.

Teaching Assistant

CS471 Graph Machine Learning and Mining	Spring 2023, 2024 and 2025	*KAIST*
CS376 Machine Learning	Fall 2021, 2022, 2023, 2024 and 2025	*KAIST*
CS665(DS532) Advanced Data Mining	Fall 2023	*KAIST*
CS492 Special Topics in Computer Science: Graph Machine Learning and Mining	Spring 2022	*KAIST*

Professional Services

Peer Review

Conference Reviewer: NeurIPS (Top Reviewer 2026), ICML (Gold Reviewer 2026), KDD, AAAI, ACL ARR

Journal Reviewer: TKDD, TMLR